SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4303
GRADED MARKS – 92/100

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

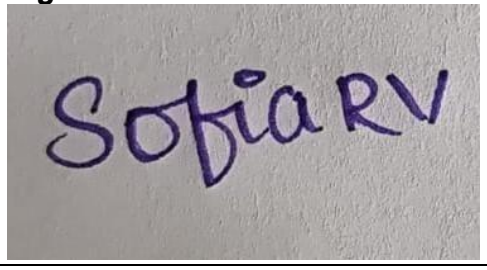
Weightage: 10%

QUESTIONS: **Total Marks: 10**

Code of Conduct:

I SOFIA RV – 192511055 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- **The team needs to embed a promotional video on the homepage.**
- **A registration form must securely send user data to the server.**
- **The navigation menu must follow semantic HTML standards.**
- **XML configuration files are being used but contain syntax errors.**
- **The application must communicate over the correct web protocol.**

Data Snippets Provided

Snippet 1: Video Embedding Options

A. **<media src="video.mp4"></media>**
B. **<video src="video.mp4" controls></video>**
C. **<movie src="video.mp4"></movie>**
D. **<vid src="video.mp4"></vid>**

Snippet 2: Form Submission Methods

A. **GET**
B. **POST**
C. **PUT**
D. **DELETE**

Snippet 3: Navigation Elements

A. ** **
B. **<nav>**
C. **<section>**
D. **<header>**

Snippet 4: XML Syntax

A. **<note><to>User</to><from>Admin</from>**
B. **<note><to>User</to><from>Admin</from></note>**
C. **<note><to>User</from></note>**
D. **<note to="User"></note**

Snippet 5: Web Communication Protocols

A. **FTP**
B. **HTTP**
C. **SMTP**
D. **SNMP**

Questions:

1. **Select the correct HTML tag for video embedding and justify your answer.**
2. **Identify the appropriate HTTP method for form submission and explain why.**
3. **Choose the proper semantic element for navigation and justify its usage.**
4. **Identify the correct XML syntax and explain the error in other options.**
5. **Choose the correct protocol for web communication and justify your choice.**

1. Correct HTML Tag for Video Embedding

Correct Answer: Option B — <video src="video.mp4" controls></video>

Explanation:

The HTML5 <video> element is used to embed video content directly into a web page. It allows users to watch videos without requiring external plugins.

- The src attribute specifies the location of the video file.
- The controls attribute displays playback controls such as play, pause, volume, and fullscreen.
- The <video> element supports commonly used video formats, depending on browser compatibility.

Why the other options are incorrect:

- A. <media>: Not a standard HTML element for embedding video.
- C. <movie>: Not a standard HTML element.

- D. <vid>: Not a standard HTML element.

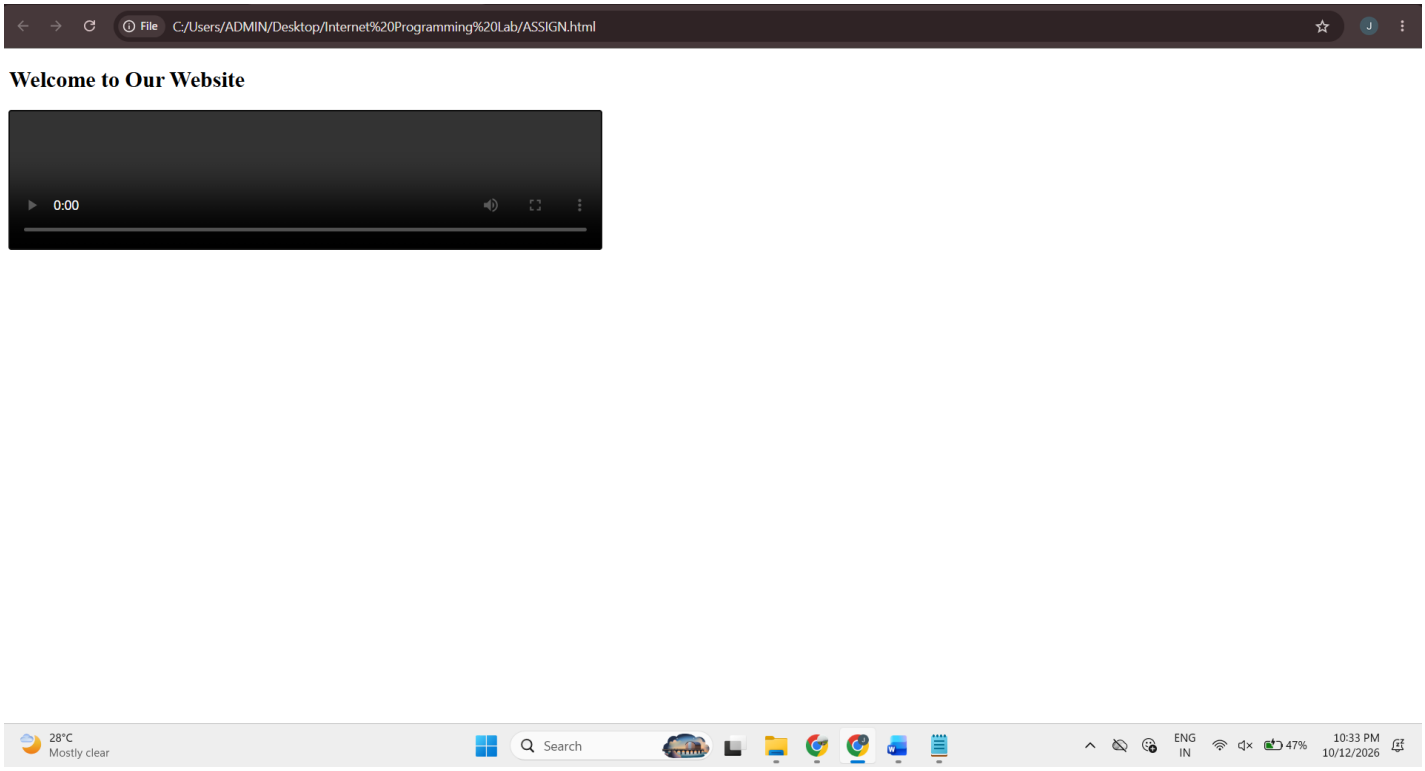
Application to the case: The startup should use the <video> element to embed its promotional video on the homepage, providing a user-friendly and standards-compliant solution.

Example:

<video src="video.mp4" controls width="640">

Your browser does not support video playback.

</video>



2. Appropriate HTTP Method for Form Submission

Correct Answer: Option B — POST

Explanation:

The HTTP POST method is appropriate for submitting registration data to a server because it sends the submitted information in the request body rather than placing it in the URL.

Advantages of using POST include:

- Privacy: Form data is not displayed directly in the URL.
- Data handling: It supports submitting larger amounts of form data than is practical with GET URLs.
- Registration processing: It is commonly used to create user accounts and submit information that changes server-side data.

However, POST alone does not guarantee security. The application must use HTTPS, validate inputs on the

server, and securely handle passwords and other sensitive information.

Why the other options are incorrect:

- A. GET: Primarily retrieves data and may expose submitted values in the URL.
- C. PUT: Commonly used to create or replace a resource at a specified URI; it is not the usual choice for a standard registration form.
- D. DELETE: Used to delete a resource.

Example:

```
<form action="/register" method="POST">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required>

    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required>

    <button type="submit">Register</button>
</form>
```

Application to the case: The startup should use POST to submit registration details and HTTPS to protect data during transmission.



3. Proper Semantic Element for Navigation

Correct Answer: Option B — <nav>

Explanation:

The HTML5 <nav> element represents a section of a web page containing major navigation links, such as links to Home, About Us, Services, and Contact.

Its advantages include:

- Semantic meaning: Clearly identifies the navigation section.
- Accessibility: Helps screen readers identify navigation landmarks.
- Maintainability: Makes the structure of the web page easier for developers to understand.
- SEO and structure: Provides meaningful document structure for browsers and search engines.

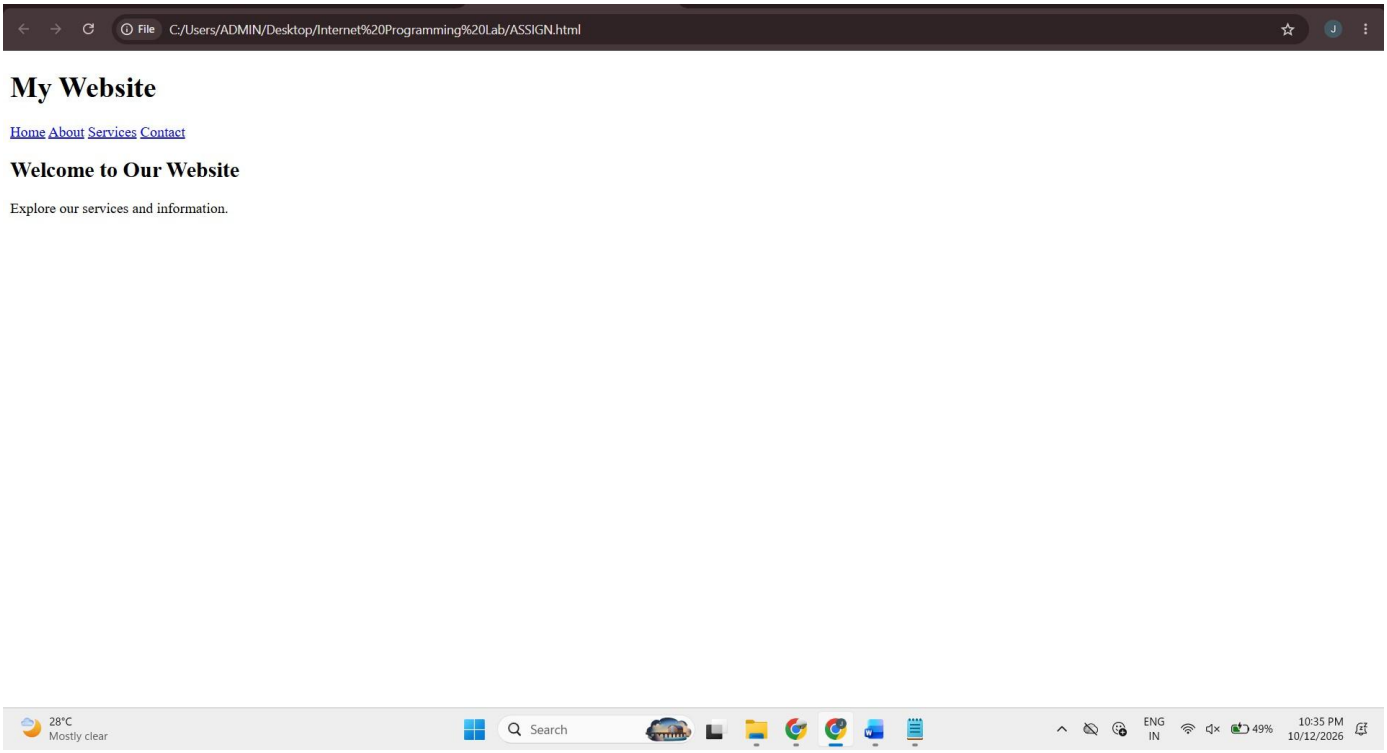
Why the other options are incorrect:

- A. : A generic container without inherent semantic meaning.
- C. <section>: Represents a thematic grouping of content, not specifically navigation.
- D. <header>: Represents introductory content or a header area and may contain navigation, but is not itself the dedicated navigation element.

Example:

```
<nav>
  <a href="/">Home</a>
  <a href="/about">About Us</a>
  <a href="/services">Services</a>
  <a href="/contact">Contact</a>
</nav>
```

Application to the case: Using <nav> ensures the startup's navigation menu follows semantic HTML standards and improves accessibility and code clarity.



4. Correct XML Syntax and Errors in Other Options

Correct Answer: Option B — <note><to>User</to><from>Admin</from></note>

Explanation:

XML (Extensible Markup Language) is used to store and transport structured data. XML documents must follow strict syntax rules to be well-formed.

Option B is correct because:

- It contains a single root element, <note>.
- Every opening tag has a matching closing tag.
- The elements are properly nested.
- The document has a valid overall structure.

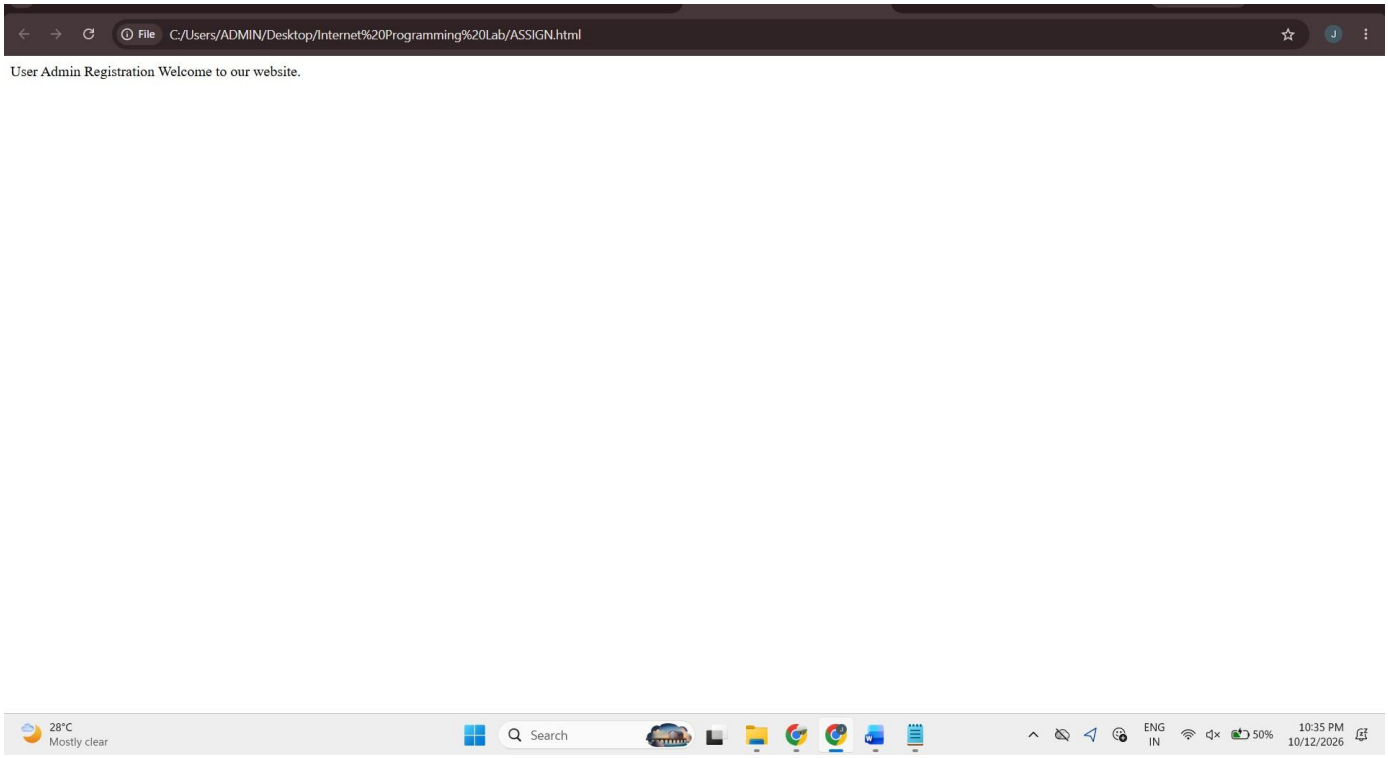
Analysis of the other options:

Option	Error
A	The root element <note> is not closed, making the document incomplete.
B	Correct: all elements are properly opened, nested, and closed.
C	The <to> element is incorrectly closed with </from>, causing a tag mismatch.
D	The <note> opening tag is malformed because its closing angle bracket > is missing, and the element is not properly closed.

Example of the correct XML:

```
<?xml version="1.0" encoding="UTF-8"?>
<note>
    <to>User</to>
    <from>Admin</from>
</note>
```

Application to the case: The startup should validate its XML configuration files using an XML parser to identify mismatched tags, missing closing tags, and other well-formedness errors before deploying the application.



5. Correct Protocol for Web Communication

Correct Answer: Option B — HTTP (Hypertext Transfer Protocol)

Explanation:

HTTP is an application-layer protocol used for communication between web clients, such as browsers, and web servers. It defines how requests and responses are exchanged to retrieve web pages, submit forms, and access web resources.

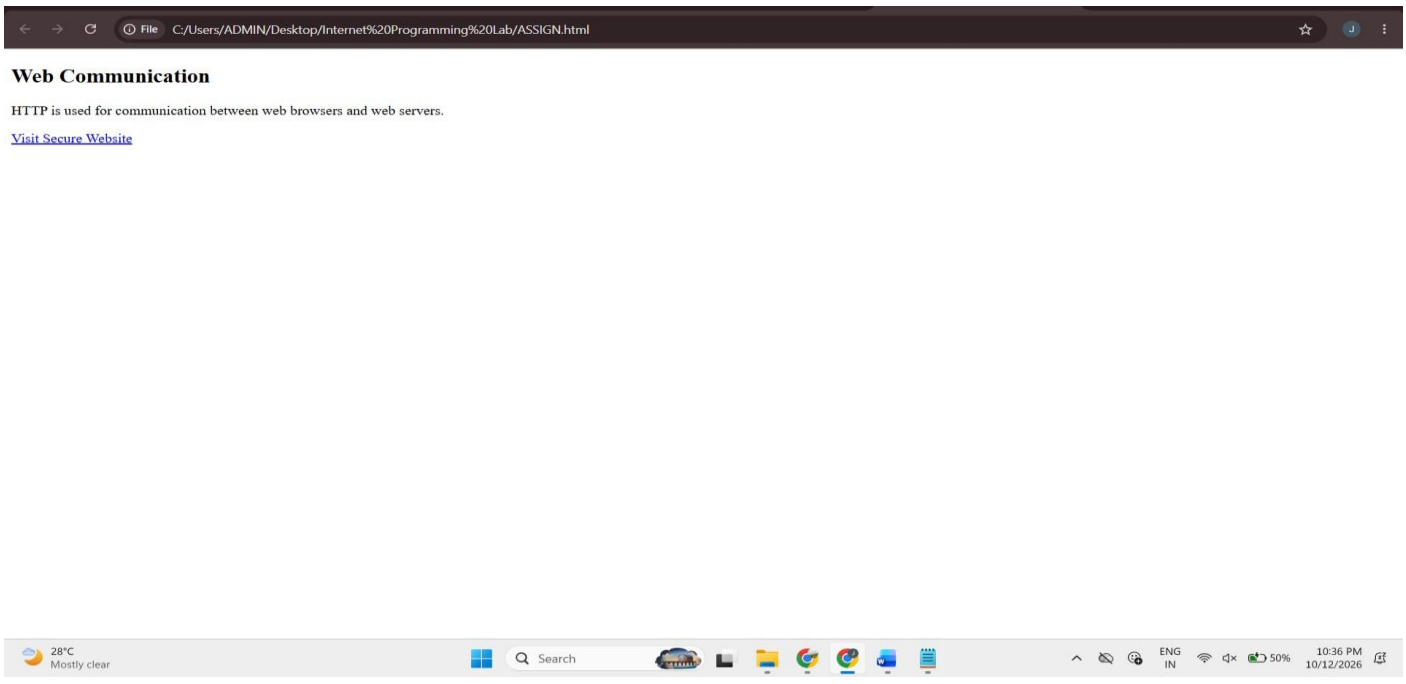
For example, when a user opens the startup's website, the browser sends an HTTP request and the server returns an HTTP response.

Why the other options are incorrect:

- A. FTP (File Transfer Protocol): Used primarily to transfer files between systems.
- B. HTTP: Used for communication between web browsers and web servers.
- C. SMTP (Simple Mail Transfer Protocol): Used to send email messages.
- D. SNMP (Simple Network Management Protocol): Used to monitor and manage network devices.

Important security consideration: For a modern web application, the startup should use HTTPS (HTTP Secure) rather than unencrypted HTTP. HTTPS uses TLS to protect data in transit, authenticate the server, and help prevent interception or modification of communications.

Application to the case: The application should communicate using HTTPS for secure delivery of web pages and registration data.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Mark Awarded
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	23/25
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	23/25
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	18/20
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	19/20
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	9/10
Total	**100**				**Overall Mark**	**92/100**

Total Score: 92/100
Grade: Level 4 – Exemplary